THE EXTENDED ALTERNATING CECH COMPLEX OF A LOCAL COMPLETE INTERSECTION RESTRICTS TO AN ACYCLIC COMPLEX OFF THE CLOSED SET

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Editorial note. No omission markers in-span. The post-proof remark (the lemma holds for any extended alternating Cech complex from a finite open covering) is context, not an obligation.

Allowed standard prerequisites (editorial).

- Local description of the extended alternating complex. Remark remark-support-c-equations (same chapter, structural). Fixes the covering U_i = D(f_i) of X - Z and the complex of equation-extended-alternating.
- Alternating Cech complexes are trivial over a member. Cohomology, lemma alternating-cech-trivial (cross-chapter, named). The alternating Cech complex of a covering restricted to an open contained in one member is homotopic to zero.

Source reference. Original reference 0G6T: focused-seeds/geometry/S9-perf2-extended-alternating-zero/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 01FG: focused-seeds/geometry/S9-perf2-extended-alternating-zero/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original equation 0G7J: sources/stacks-perfect/pinned-original-source.tex, source lines 1107–1107. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original remark 0G7I: sources/stacks-perfect/pinned-original-source.tex, source lines 1093–1093. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *With X, $f_1, \ldots, f_c \in \Gamma(X, \mathcal{O}_X)$, and $\mathcal{F}$ as in Remark 0G7I the complex (0G7J) restricts to an acyclic complex over $X \setminus Z$.*

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Proof. Let $W \subset X \setminus Z$ be an open subset. Evaluating the complex of sheaves (0G7J) on W we obtain the complex

$$\mathcal{F}(W) \to \bigoplus\nolimits_{i_0} \mathcal{F}(U_{i_0} \cap W) \to \bigoplus\nolimits_{i_0 < i_1} \mathcal{F}(U_{i_0 i_1} \cap W) \to \ldots$$

In other words, we obtain the extended ordered Čech complex for the covering $W = \bigcup U_i \cap W$ and the standard ordering on $\{1, \ldots, c\}$, see Cohomology, Section 01FG. By Cohomology, Lemma 0G6T this complex is homotopic to zero as soon as W is contained in $V(f_i)$ for some $1 \leq i \leq c$. This finishes the proof. $\square$